# Summary

Erik An

*<2026-02-02 Mon>*

## Contents

**1 CHEAT SHEET: Real Analysis & Calc** **2**

1.1 1. DAILY THEOREMS . . . . . 2

1.2 2. SEQUENCES & SERIES . . . . . 2

1.3 3. CONTINUITY & METRIC SPACES . . . . . 2

1.4 4. DERIVATIVES & APPROX . . . . . 2

1.5 5. INTEGRATION . . . . . 3

# 1 CHEAT SHEET: Real Analysis & Calc

## 1.1 1. DAILY THEOREMS

- **BW:** Bounded seq in $\mathbb{R}^n$ has convergent subseq.
- **IVT:** $f$ cont on $[a, b]$, $y$ betw $f(a), f(b) \implies \exists c, f(c) = y$.
- **MVT:** Cont $[a, b]$, diff $(a, b) \implies \exists c, f'(c) = \frac{f(b)-f(a)}{b-a}$.
- **Continuity ($\epsilon - \delta$):** $\forall \epsilon > 0, \exists \delta > 0 : |x - c| < \delta \implies |f(x) - f(c)| < \epsilon$.
- **$\pi$:** Review precise definition.

## 1.2 2. SEQUENCES & SERIES

- **Recursive Limits:**
  1. Prove Bounded (Induction).
  2. Prove Monotonic ($x_{n+1} \leq x_n$).
  3. Solve fixed point $L = f(L)$ (e.g., $L = \frac{1}{2}(L + 2/L)$).
- **Convergence:** Alt. Harmonic Series converges.
- **Zero Limits:** Prove $\lim \frac{(-1)^n}{n} = 0$.
- **Euler:** $(1 + \frac{1}{n})^n \to e$.

## 1.3 3. CONTINUITY & METRIC SPACES

- **Metric:** $d(x, y) \geq 0$, symm, $d(x, y) = 0 \iff x = y$, $\Delta$ -ineq ($d(xz) \leq d(xy) + d(yz)$).
- **$\mathbb{R}^2$:** Convergence in Taxi ($d_1$) $\iff$ Euclidean ($d_2$).
- **Piecewise:** Check $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^-} f(x) = f(0)$ (e.g., $e^{-1/x}$).
- **Comp:** $f, g$ cont $\implies f \circ g$ cont.

## 1.4 4. DERIVATIVES & APPROX

- **The Trick:** $\frac{d}{dx} f(x)^{g(x)} \implies \frac{d}{dx} \exp(g(x) \ln f(x))$.
- **Must Know:** $e^{3x+1}, \ln(5x^2), \sqrt{1 + e^{2x}}$.
- **Def:** $f'(x_0) = \lim_{x\to x_0} \frac{f(x)-f(x_0)}{x-x_0}$ (Use for prod rule at 0).
- **Newton's:** $x_{new} = x_{old} - \frac{f(x_{old})}{f'(x_{old})}$.
- **Taylor:** $P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x - a)^k$.

## 1.5 5. INTEGRATION

- **Sums:** $L(f,P) = \sum m_i \Delta x_i \leq \int \leq U(f,P) = \sum M_i \Delta x_i$.
- **Improper:** $\int_0^1 x^s dx$ conv $\iff s > -1$.
- **Parts:** $\int u dv = uv - \int v du$ (for $xe^x$).
- **Subst:** $\int f(g(x))g'(x)dx = \int f(u)du$ (for $2xe^{x^2}$).